

“Meet the Deer King”: “Splash-Ink” Interaction in the Innovative VR Game Based on Dunhuang Art and Culture

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ABSTRACT

VR technologies enable 3D visualization and 3D interaction modalities, allowing the general public to get user-centered experience and explore an immersive realistic cultural environment. The art and culture of Dunhuang grotto murals is the world’s art treasure. According to the traditional story of “Deer King” and the knowledges of preservation and restoration of the murals, we designed “Meet the Deer King”, a VR game. In this game, we explored and designed “splash-ink” interaction, which derives from traditional Chinese painting style, and we applied this interaction in different scenes. The information about the original game, including its creation background, research focus, game content and implementation effects are mainly described in the supplementary materials.

CCS CONCEPTS

• **Human-centered computing** → Virtual reality; • **Applied computing** → Arts and humanities; Performing arts.

KEYWORDS

Dunhuang art, the Deer King jataka story, virtual reality, “splash-ink” interaction

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1 INTRODUCTION

“Meet the Deer King” was originally designed as a VR game of the art and culture of Dunhuang grotto murals. The feature of the whole game is to create an interactive mechanism by borrowing the form of splash-ink from traditional Chinese painting techniques. Compared with the existing VR experience, VR game increases the interest and immersion of interaction. Later designers found that player interactions, with high quality and high feedback, can not only experience the operability of the game world, but also influence their thinking [Qian and Xu, 2021]. Therefore this game is set as a prototype to experiment the “splash-ink” interaction mechanism. while remaining the feature of Dunhuang traditional art and its restoration method.

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2 THE “SPLASH-INK” INTERACTION

The main interaction design of the whole game is the “splash-ink” interaction, which comes from the Chinese painting style for Dunhuang mural. It is characterized by imprecision, for a lot of drops created at the same time, while more interesting.

2.1 The trigger and input

Hold the brush in the virtual environment, aim the tip at the target you want to splash and quickly sweep the brush sideways. Then hold the Oculus Touch controller with the virtual paintbrush, point the top of the controller in the direction of the target, and sweep the controller in any lateral direction. There are no physical buttons to press during the process.

2.2 The visual effects

The scene, which can be “drawn” by the ink drops, is rendered using 2D Sprite. Before being hit by the ink drop, the mountain is rendered in a monochromatic translucent style using a self-developed shader. After being hit, a sequence of frames is animated to achieve the effect of the mountain being splashed. The ink drops are implemented using the unity engine’s particle system, which uses its built-in physics functions to implement collision detection and collision event notification for ink drops

2.3 The method of motion recognition

After obtaining the controller movement speed, the velocity vector in its world space is transformed into the local space of the joystick game object, and then the two right-angle velocity vectors perpendicular to the up-and-down direction of the controller are merged. The merged velocity direction is the velocity of the horizontal sweep of the controller. Using this velocity as the basis for judgment, when the velocity exceeds the judgment threshold, the ink drops start to be emitted. Here are the formulas to tell whether fire the particles and the particle speed.

$$K = V_r / V_{th} V_p = \max(2.2, k) * s$$

Let V_r be the initial speed at which the player throws the pen. Let k be the Percentage values. V_{th} is the minimal speed of fire. Then the value of k can be figured out. Then set initial firing speed of the ink particles is V_p , the result of V_p can be shown. If k is smaller than 2.2, than set the figure $\max(2.2, k)$ as 2.2 to insure a basic particle speed. Therefore players can make good particle fire with not much effort.

3 “SPLASH-INK” INTERACTIONS IN DIFFERENT LEVELS

At present, many experts at home and abroad have developed a variety of games on the Unity platform. For example, Wu Chuanmin

and others have completed the design and development of first-person shooter games based on Unity3D [Zhang et al., 2014]. There are three implementation forms designed for the "splash-ink" interaction: UI interaction, engagement interaction and interaction with the scene. All of them are applied into the games' three parts: multi-line narration, engagement and restoration.

3.1 UI interaction

UI interaction is all about getting the player to interact with the UI through splashing ink. This interaction is not accurate because the droplets' number sprayed at the same time is large and the direction is not consistent. However, by recording the number of ink drops touched by different UI interfaces over a period of time, it is possible to compare which UI has been sprayed the most, and thus determine which UI the player has chosen. At the same time, quantitative ink splashes before confirming and triggering the UI, confirming that can prevent accidental contact.

3.2 Engagement interaction

In the engagement part, players use "splash-ink" interaction as the weapon to attack and destroy the enemies— the soldiers of the king. When the soldier are hit by the ink, they will hurt and flash red and stop for a few seconds to show clear feedback of injury, adding a sense of control to the interaction. After three times' attack, the soldier can be destroyed.

As the number of monster waves increases and the turret stats increase, the number of monsters, stats, and various special abilities will also increase [Shen, 2019. Feng, 2019.]

3.3 Interaction with the scene

In a relatively fixed scene, the main function of splash-ink interaction is to guide users into the scene from the blank and promote the development of narrative. And combined with the narrative, reflected in different situations in different values.

In the multi-line narration part, players are intended to participate in the story plot and feel the empathy and intersection with the Deer King. Players need to paint the mountains to start the narrative story and the river bank to help the Deer King to save people. so as to deepen the experience from the basic level of operation to the emotional level.

Although the "splash-ink" interaction is not as accurate as the ordinary interactive triggering mode, such as firing bullets and teleport, this interaction is also in line with the normal cognition of people. The vivid-ness of the action increases the fun of the interaction, and the Oriental charm contained in the ink itself can be released to the maximum extent. In the feel of the design, through the formula to measure the waving speed, and then calculate the speed multiplied by the rate, to judge whether splashing ink and the effect of splashing ink, the process is controllable, easy to master. At the same time, the systematic development of interactive mechanics makes it possible for any developer to quickly use this set of mechanics in the future, making possible that new game designs are based on this set of gameplay.

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